TUNIS BUSINESS SCHOOL

Chapter 2

Project Scheduling

Potential method, probabilistic networks and cost minimisation

Production & Operations Management

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CHAPTER CONTENTS

What we will cover

Introduction — modelling a project	Activities · precedence · two network methods
1. Scheduling with the potential method	Network · earliest / latest dates · floats
2. The probabilistic network	Three-point estimates · normal law · risk
3. Cost minimisation	Crashing · direct vs indirect costs · optimum
Exercise after every section + detailed solutions	Every network drawn · every calculation shown



CHAPTER OBJECTIVE

Chapter objective

A crucial problem in completing a project is to determine in which order the different activities must follow one another, given their precedence constraints, so as to minimise the total execution time of the project.

> ***We build the execution timetable of the activities: when each activity can start at the earliest, at the latest, and which activities must not be delayed.***

INTRODUCTION

Two ways of modelling a project network

In large projects — a ship, an aircraft, a building — the key question is the execution timetable. Both methods below represent the project as a graph (a network); they give the same dates.

POTENTIAL METHOD (MPM)

France, 1958 (B. Roy)

Activity = NODE. Precedence = ARC.



PERT METHOD

United States, 1958

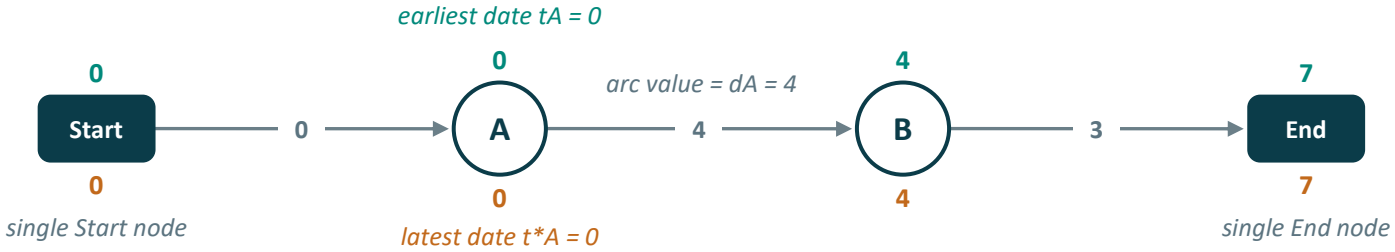
Activity = ARC. Events = nodes; may need dummy arcs.



In this course we use the potential method: it is drawn systematically from the data table and never needs dummy arcs.

1. SCHEDULING BY THE POTENTIAL METHOD

The building blocks of a potential network



Node = activity

Each activity is drawn ONCE, as a circle with its name. Start and End are two extra nodes of zero duration.

Arc = precedence

An arrow i → j means: j cannot start before i is finished. The arc carries the duration di of the activity it LEAVES. Its length means nothing.

Dates on the node

Earliest start ti is written ABOVE the node (green). Latest start t*i is written BELOW the node (orange).

1. SCHEDULING BY THE POTENTIAL METHOD

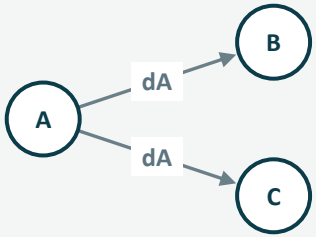
Four basic configurations to recognise

Succession



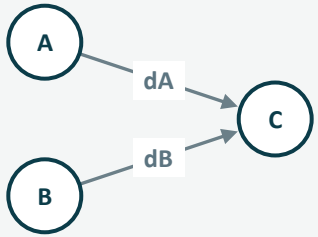
B follows A: B starts when A is finished.

Divergence



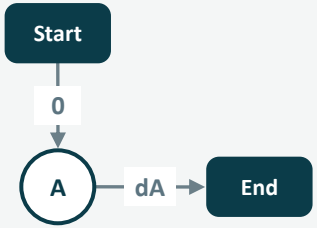
B and C both need A: they can start together when A is finished.

Convergence



C needs A AND B: it waits for the LAST of the two to finish.

Start and End



Activities without predecessor leave Start (arc 0); activities without successor go to End.

1. SCHEDULING BY THE POTENTIAL METHOD

How to draw a potential network — the method

01 **List the successors.** From the table of predecessors, write for each activity the activities that follow it.

02 **Give each activity a level.** Level 1 = no predecessor. Level k = all predecessors are on lower levels. Each level is one column.

03 **Place the nodes column by column,** from left to right, and draw one arrow per precedence. Each arrow carries the duration of the activity it leaves.

04 **Close the network.** Link level-1 activities to Start (arc 0) and activities without successor to End (arc = their duration).

05 **Check.** Every activity appears once; every arrow goes left → right; no cycle; one Start, one End.

Avoid crossings by placing on the same row the activities that follow each other.

1. EXAMPLE 1 — A NEW BUILDING

The data: activities and precedences

N°	Activity	Duration (days)	Predecessors
1	Earthworks	5	—
2	Foundations	4	1
3	Load-bearing columns	2	2
4	Roof framework	2	3
5	Roofing	3	4
6	Masonry	5	3
7	Plumbing and electrical work	3	3
8	Concrete slab pouring	3	3
9	Heating	4	6, 7, 8
10	Plastering	10	5, 9
11	Finishing works	5	10

QUESTIONS

1. Draw the potential network.
2. Compute the earliest start dates.
3. Compute the latest start dates.
4. Compute the floats in a table.
5. Critical path and minimum duration.

> ***Objective: erect the building in the minimum total time.***

EXAMPLE 1 — STEP 1

Successors and levels

Activity	1	2	3	4	5	6	7	8	9	10	11
Predecessors	—	1	2	3	4	3	3	3	6,7,8	5,9	10
Successors	2	3	4,6,7,8	5	10	9	9	9	10	11	End
Level	1	2	3	4	5	4	4	4	5	6	7

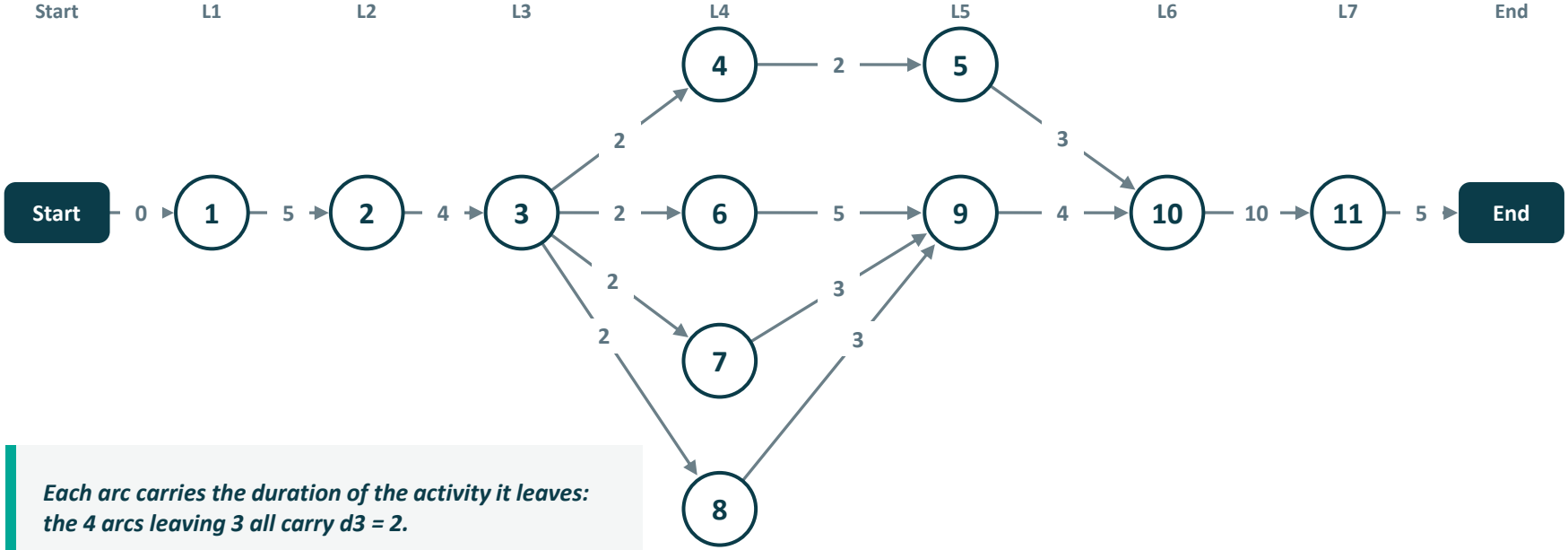
L1 **Level 1:** activity 1 (no predecessor) → linked to Start.

L4 **Level 4:** 4, 6, 7 and 8 all follow activity 3 → they form one column (divergence).

L6 **Level 6:** 10 waits for 5 (level 5) AND 9 (level 5) → convergence. 11 (level 7) is linked to End.

EXAMPLE 1 — STEP 2

Drawing the potential network



Each arc carries the duration of the activity it leaves: the 4 arcs leaving 3 all carry d3 = 2.

1.1 EARLIEST START DATES

The forward pass: from Start to End

The earliest start date ti is the first date at which activity i can start, if every activity before it starts as early as possible. We go FORWARD, from left to right, adding the durations.

FORWARD PASS

$$tj = max\,(\,ti + di\,)$$
$$over\ all\ predecessors\ i\ of\ j$$

1. **Start:** tStart = 0, so every activity without predecessor has ti = 0.
2. **One predecessor:** tj = ti + di (simple addition).
3. **Several predecessors (convergence):** take the MAXIMUM — j waits for the last one.
4. **End:** tEnd = minimum total duration of the project.

EXAMPLE 1 — EARLIEST DATES

Forward pass: detailed calculation

Activity	Predecessors	Formula (max at a convergence)	Numerical application	Earliest ti
1	—	t1 = 0	No predecessor: linked to Start	**0**
2	1	t2 = t1 + d1	0 + 5	**5**
3	2	t3 = t2 + d2	5 + 4	**9**
4	3	t4 = t3 + d3	9 + 2	**11**
5	4	t5 = t4 + d4	11 + 2	**13**
6	3	t6 = t3 + d3	9 + 2	**11**
7	3	t7 = t3 + d3	9 + 2	**11**
8	3	t8 = t3 + d3	9 + 2	**11**
9	6, 7, 8	t9 = max(t6+d6 ; t7+d7 ; t8+d8)	max(11 + 5 ; 11 + 3 ; 11 + 3) = max(16 ; 14 ; 14)	**16**
10	5, 9	t10 = max(t5+d5 ; t9+d9)	max(13 + 3 ; 16 + 4) = max(16 ; 20)	**20**
11	10	t11 = t10 + d10	20 + 10	**30**
End	11	tEnd = t11 + d11	30 + 5	**35**

1.2 LATEST START DATES

The backward pass: from End to Start

The latest start date t*i is the last date at which activity i can start without delaying the end of the project. We go BACKWARD, from right to left, subtracting the durations.

BACKWARD PASS

$$t^*i = \min (t^*j) - di$$
$$\text{over all successors } j \text{ of } i$$

1. **End:** t*End = tEnd (we keep the minimum duration found by the forward pass).
2. **One successor:** t*i = t*j – di (simple subtraction).
3. **Several successors (divergence):** take the MINIMUM of the t*j, then subtract di.
4. **Check:** we must find t*Start = 0 — otherwise there is a mistake.

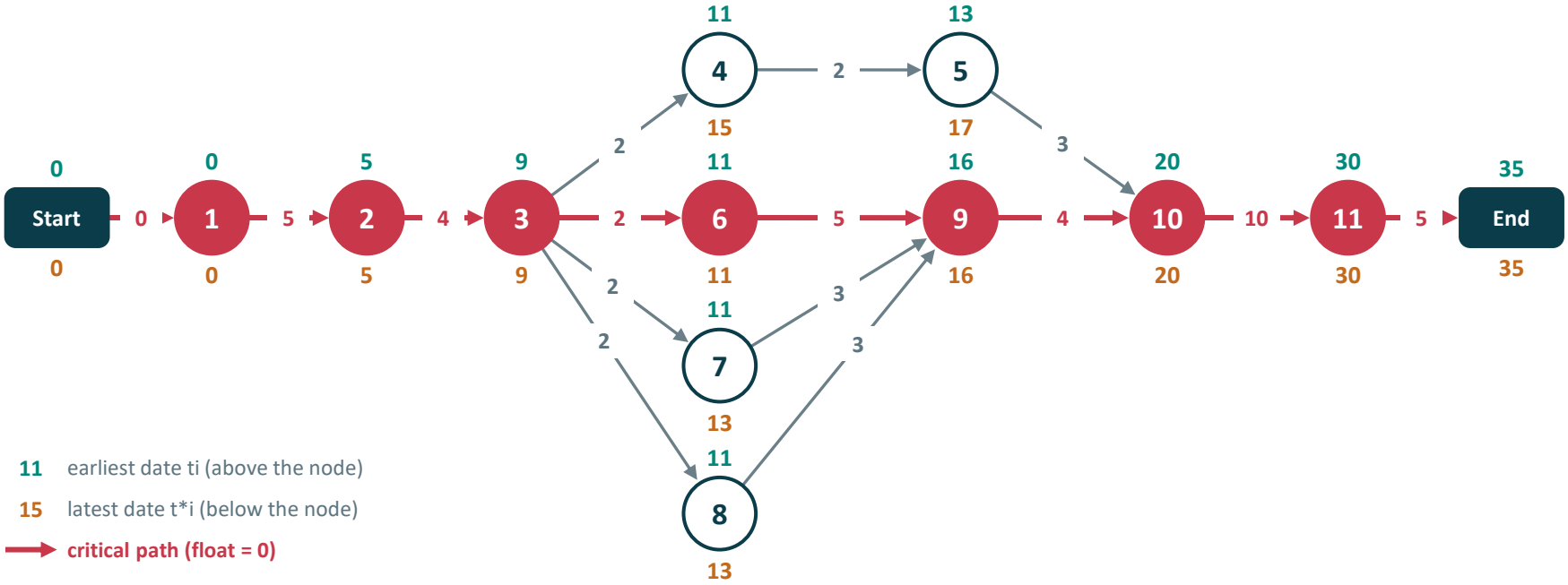
EXAMPLE 1 — LATEST DATES

Backward pass: detailed calculation

Activity	Successors	Formula (min at a divergence)	Numerical application	Latest t*i
End	—	t*End = tEnd	Project duration (end of the forward pass)	**35**
11	End	t*11 = t*End − d11	35 − 5	**30**
10	11	t*10 = t*11 − d10	30 − 10	**20**
9	10	t*9 = t*10 − d9	20 − 4	**16**
8	9	t*8 = t*9 − d8	16 − 3	**13**
7	9	t*7 = t*9 − d7	16 − 3	**13**
6	9	t*6 = t*9 − d6	16 − 5	**11**
5	10	t*5 = t*10 − d5	20 − 3	**17**
4	5	t*4 = t*5 − d4	17 − 2	**15**
3	4, 6, 7, 8	t*3 = min(t*4 ; t*6 ; t*7 ; t*8) − d3	min(15 ; 11 ; 13 ; 13) − 2 = 11 − 2	**9**
2	3	t*2 = t*3 − d2	9 − 4	**5**
1	2	t*1 = t*2 − d1	5 − 5	**0**

EXAMPLE 1 — RESULT

The potential network with all dates



1.3 FLOATS AND CRITICAL PATH

Identifying the activities to watch

TOTAL FLOAT OF AN ACTIVITY

$$mi = t^*i - ti$$

The float is the delay an activity can take without changing the minimum total duration of the project.

CRITICAL activities

Float = 0. Together they form the critical path, the longest path from Start to End. Any delay on them delays the whole project: they are monitored first.

NON-CRITICAL activities

Float > 0. They can be delayed within their float. Shortening them is useless: it only increases their float, not the project's speed.

EXAMPLE 1 — FLOATS

Floats table: latest – earliest

Activity	1	2	3	4	5	6	7	8	9	10	11
Latest start t*i	0	5	9	15	17	11	13	13	16	20	30
Earliest start ti	0	5	9	11	13	11	11	11	16	20	30
Float mi = t*i – ti	0	0	0	4	4	0	2	2	0	0	0
Critical?	Yes	Yes	Yes	No	No	Yes	No	No	Yes	Yes	Yes

CRITICAL PATH Start – 1 – 2 – 3 – 6 – 9 – 10 – 11 – End = 5 + 4 + 2 + 5 + 4 + 10 + 5 = 35 days

*Example: roof framework (4): m4 = t*4 – t4 = 15 – 11 = 4 days. It can be delayed 4 days without changing the 35-day duration.*

EXERCISE 1 · POTENTIAL METHOD

Graph, dates, floats and critical path

A project is made of eight activities. Durations are given in weeks.

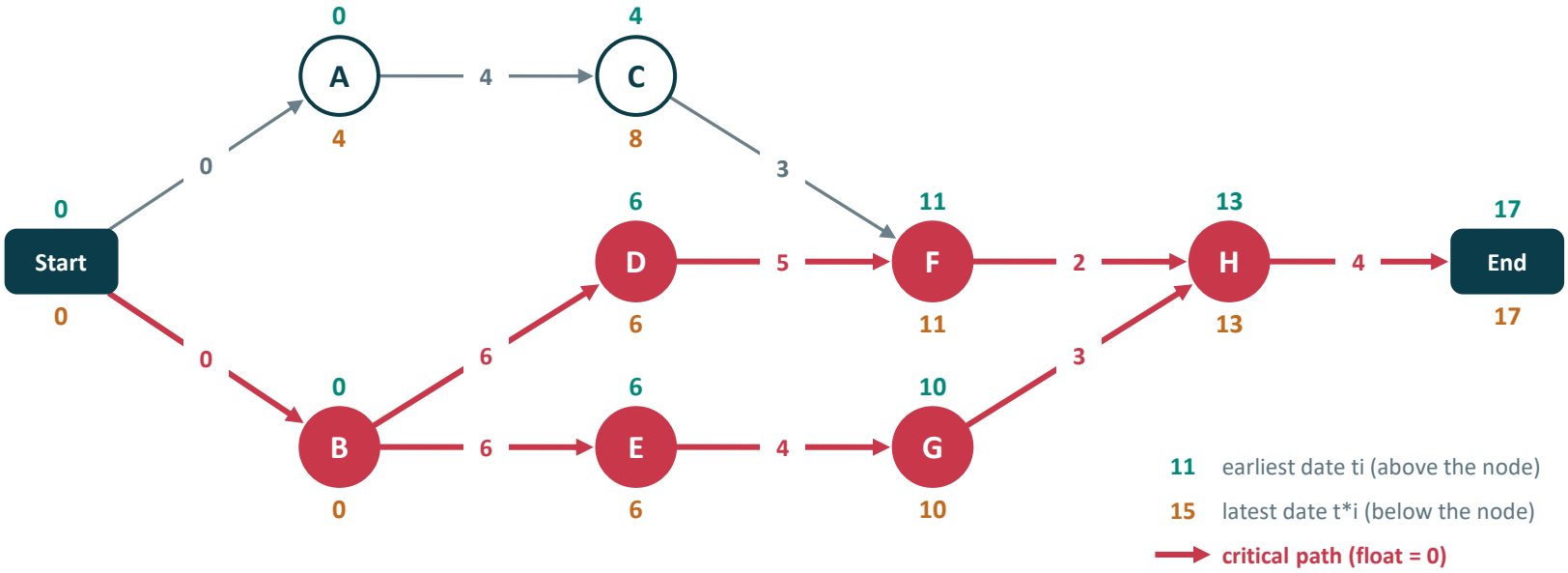
Activity	Duration (weeks)	Predecessors
A	4	—
B	6	—
C	3	A
D	5	B
E	4	B
F	2	C, D
G	3	E
H	4	F, G

QUESTIONS

1. Draw the potential network.
2. Compute the earliest dates (forward pass).
3. Compute the latest dates (backward pass).
4. Compute the floats in a table.
5. Give the critical path(s) and the minimum duration.

EXERCISE 1 · SOLUTION

The potential network with all dates



EXERCISE 1 · SOLUTION

Earliest dates: detailed calculation

Activity	Predecessors	Formula (max at a convergence)	Numerical application	Earliest ti
A	—	tA = 0	No predecessor: linked to Start	**0**
B	—	tB = 0	No predecessor: linked to Start	**0**
C	A	tC = tA + dA	0 + 4	**4**
D	B	tD = tB + dB	0 + 6	**6**
E	B	tE = tB + dB	0 + 6	**6**
F	C, D	tF = max(tC+dC ; tD+dD)	max(4 + 3 ; 6 + 5) = max(7 ; 11)	**11**
G	E	tG = tE + dE	6 + 4	**10**
H	F, G	tH = max(tF+dF ; tG+dG)	max(11 + 2 ; 10 + 3) = max(13 ; 13)	**13**
End	H	tEnd = tH + dH	13 + 4	**17**

EXERCISE 1 · SOLUTION

Latest dates: detailed calculation

Activity	Successors	Formula (min at a divergence)	Numerical application	Latest t*i
End	—	t*End = tEnd	Project duration (end of the forward pass)	**17**
H	End	t*H = t*End − dH	17 − 4	**13**
G	H	t*G = t*H − dG	13 − 3	**10**
F	H	t*F = t*H − dF	13 − 2	**11**
E	G	t*E = t*G − dE	10 − 4	**6**
D	F	t*D = t*F − dD	11 − 5	**6**
C	F	t*C = t*F − dC	11 − 3	**8**
B	D, E	t*B = min(t*D ; t*E) − dB	min(6 ; 6) − 6 = 6 − 6	**0**
A	C	t*A = t*C − dA	8 − 4	**4**

EXERCISE 1 · SOLUTION

Floats table and critical paths

Activity	A	B	C	D	E	F	G	H
Latest start t*i	4	0	8	6	6	11	10	13
Earliest start ti	0	0	4	6	6	11	10	13
Float mi = t*i – ti	4	0	4	0	0	0	0	0
Critical?	No	Yes	No	Yes	Yes	Yes	Yes	Yes

TWO CRITICAL PATHS B–D–F–H = 6+5+2+4 = 17 · B–E–G–H = 6+4+3+4 = 17 weeks

NEXT → To shorten this project, BOTH critical paths must be shortened together — Section 3.

2. THE PROBABILISTIC NETWORK

When durations are uncertain

Until now durations were known with certainty. In reality they vary, especially for activities never done before. Managers then give three estimates for each activity i:

ESTIMATED MEAN DURATION

$$ti = (ai + 4\,mi + bi) / 6$$

ESTIMATED VARIANCE

$$vi = [(bi - ai) / 6]^2$$

ai	optimistic (minimum) duration — everything goes well
mi	most likely duration — the usual case
bi	pessimistic (maximum) duration — many problems

2. THE PROBABILISTIC NETWORK

From the activities to the whole project

01 **Mean duration of the project:** μ = sum of the mean durations on the critical path = tEnd

02 **Variance of the project:** σ^2 = sum of the variances of the CRITICAL activities only; $\sigma = \sqrt{\sigma^2}$

03 **Probability law:** The project duration X follows a normal law N(μ, σ)

04 **Probability to finish within x:** $P(X \leq x) = \Phi(z)$ with $z = (x - \mu) / \sigma$ — read Φ in the standard normal table

05 **Duration for a given probability p:** find z such that $\Phi(z) = p$, then $x = \mu + z \cdot \sigma$

z	0.61	1.00	1.22	1.28	1.645	1.66	1.70	2.00	2.55
Φ(z)	0.7291	0.8413	0.8888	0.8997	0.9500	0.9515	0.9554	0.9772	0.9946

$\Phi(-z) = 1 - \Phi(z)$

2. EXAMPLE 2 — PROBABILISTIC NETWORK

The data and the questions

The activities of a project are estimated with three durations (in weeks).

Activity	a	m	b	Predecessors
A	4	8	10	—
B	2	4	6	—
C	4	7	11	—
D	2	3	5	A
E	6	8	11	B
F	1	2	3	C
G	1	3	4	D, E
H	4	5	7	F

QUESTIONS

1. Draw the potential network.
2. Estimate the mean ti and the variance vi of each activity.
3. Compute the earliest and latest dates with the mean durations; deduce the mean project duration.
4. Find the critical path and the sum of its variances.
5. P(X ≤ 17)? P(X ≤ 12)? Duration at 90 %?

EXAMPLE 2 — QUESTION 2

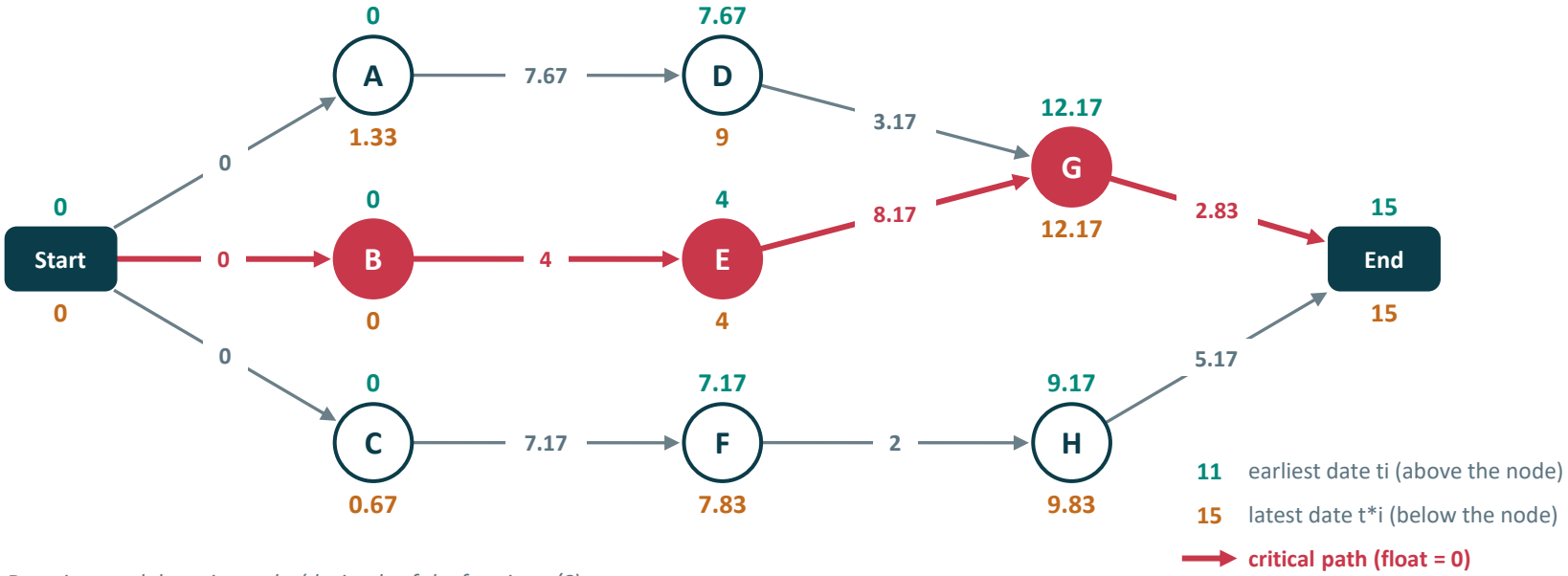
Means and variances: detailed calculation

Act.	a	m	b	ti = (a + 4m + b) / 6	ti	vi = [(b − a) / 6]²	vi
A	4	8	10	(4 + 4×8 + 10) / 6 = 46/6	7.67	[(10 − 4) / 6]² = 36/36	1.000
B	2	4	6	(2 + 4×4 + 6) / 6 = 24/6	4.00	[(6 − 2) / 6]² = 16/36	0.444
C	4	7	11	(4 + 4×7 + 11) / 6 = 43/6	7.17	[(11 − 4) / 6]² = 49/36	1.361
D	2	3	5	(2 + 4×3 + 5) / 6 = 19/6	3.17	[(5 − 2) / 6]² = 9/36	0.250
E	6	8	11	(6 + 4×8 + 11) / 6 = 49/6	8.17	[(11 − 6) / 6]² = 25/36	0.694
F	1	2	3	(1 + 4×2 + 3) / 6 = 12/6	2.00	[(3 − 1) / 6]² = 4/36	0.111
G	1	3	4	(1 + 4×3 + 4) / 6 = 17/6	2.83	[(4 − 1) / 6]² = 9/36	0.250
H	4	5	7	(4 + 4×5 + 7) / 6 = 31/6	5.17	[(7 − 4) / 6]² = 9/36	0.250

We keep the fractions in sixths (46/6, 24/6 …) for the dates: the calculation stays exact.

EXAMPLE 2 — QUESTIONS 1 AND 3

The network with the mean durations and the dates



Durations and dates in weeks (decimals of the fractions /6).

EXAMPLE 2 — QUESTION 3

Earliest dates (in sixths of a week)

Activity	Predecessors	Formula (max at a convergence)	Numerical application	Earliest ti
A	—	tA = 0	No predecessor: linked to Start	**0**
B	—	tB = 0	No predecessor: linked to Start	**0**
C	—	tC = 0	No predecessor: linked to Start	**0**
D	A	tD = tA + dA	0 + 46/6	**46/6 ≈ 7.67**
E	B	tE = tB + dB	0 + 4	**4**
F	C	tF = tC + dC	0 + 43/6	**43/6 ≈ 7.17**
G	D, E	tG = max(tD+dD ; tE+dE)	max(46/6 + 19/6 ; 4 + 49/6) = max(65/6 ; 73/6)	**73/6 ≈ 12.17**
H	F	tH = tF + dF	43/6 + 2	**55/6 ≈ 9.17**
End	G, H	tEnd = max(tG+dG ; tH+dH)	max(73/6 + 17/6 ; 55/6 + 31/6) = max(15 ; 86/6)	**15**

EXAMPLE 2 — QUESTION 3

Latest dates (in sixths of a week)

Activity	Successors	Formula (min at a divergence)	Numerical application	Latest t*i
End	—	t*End = tEnd	Project duration (end of the forward pass)	**15**
H	End	t*H = t*End − dH	15 − 31/6	**59/6 ≈ 9.83**
G	End	t*G = t*End − dG	15 − 17/6	**73/6 ≈ 12.17**
F	H	t*F = t*H − dF	59/6 − 2	**47/6 ≈ 7.83**
E	G	t*E = t*G − dE	73/6 − 49/6	**4**
D	G	t*D = t*G − dD	73/6 − 19/6	**9**
C	F	t*C = t*F − dC	47/6 − 43/6	**4/6 ≈ 0.67**
B	E	t*B = t*E − dB	4 − 4	**0**
A	D	t*A = t*D − dA	9 − 46/6	**8/6 ≈ 1.33**

EXAMPLE 2 — QUESTION 4

Floats, critical path and its variance

Activity	A	B	C	D	E	F	G	H
Latest start t*i	1.33	0	0.67	9	4	7.83	12.17	9.83
Earliest start ti	0	0	0	7.67	4	7.17	12.17	9.17
Float mi = t*i – ti	1.33	0	0.67	1.33	0	0.67	0	0.67
Critical?	No	Yes	No	No	Yes	No	Yes	No

CRITICAL PATH B – E – G

$$\mu = 24/6 + 49/6 + 17/6$$
$$= 90/6 = 15 \text{ weeks}$$

VARIANCE OF THE PROJECT

$$\sigma^2 = 16/36 + 25/36 + 9/36 = 50/36$$
$$= 1.389 \rightarrow \sigma = 1.18$$

EXAMPLE 2 — QUESTION 5

Probability of completion

$X \sim N(\mu = 15 ; \sigma = 1.18)$. $z = (x - \mu) / \sigma$, $P(X \le x) = \Phi(z)$.

Within 17 weeks	$z = (17 - 15) / 1.18 = 1.70$	$\Phi(1.70) = 0.9554$	**95.5 %**
Within 12 weeks	$z = (12 - 15) / 1.18 = -2.55$	$\Phi(-2.55) = 1 - \Phi(2.55) = 1 - 0.9946 = 0.0054$	**0.5 %**
Duration at 90 %	$\Phi(z) = 0.90 \rightarrow z = 1.28$	$x = \mu + z \cdot \sigma = 15 + 1.28 \times 1.18 = 16.51$	**16.5 weeks**

Limits: the normal law needs enough activities (central limit theorem), and only ONE critical path is analysed — a near-critical path can also delay the project.

2. EXAMPLE 3 — A CAMPUS MOBILE APP

A second probabilistic network

A student start-up develops a mobile app for the campus. Durations are estimated in weeks.

Act.	Activity	a	m	b	Pred.
A	Requirements analysis	2	3	4	—
B	User-interface design	3	5	9	A
C	Back-end development	5	8	11	A
D	Front-end development	4	6	10	B
E	Integration and testing	2	3	6	C, D
F	Pilot launch on campus	1	2	3	E

QUESTIONS

1. Means and variances of the activities.
2. Network, dates and floats; mean duration.
3. Probability to deliver within 22 weeks?
4. Probability to deliver within 19 weeks?
5. Deadline to announce with 95 % confidence?

EXAMPLE 3 — QUESTION 1

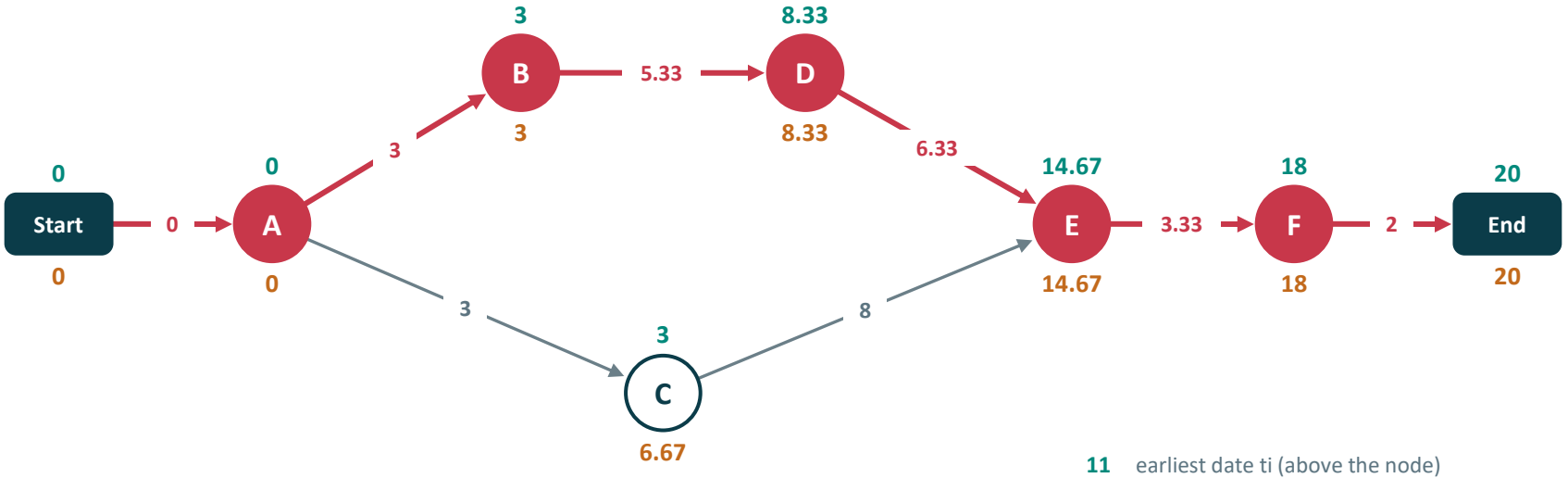
Means and variances: detailed calculation

Act.	a	m	b	ti = (a + 4m + b) / 6	ti	vi = [(b − a) / 6]²	vi
A	2	3	4	(2 + 4×3 + 4) / 6 = 18/6	3.00	[(4 − 2) / 6]² = 4/36	0.111
B	3	5	9	(3 + 4×5 + 9) / 6 = 32/6	5.33	[(9 − 3) / 6]² = 36/36	1.000
C	5	8	11	(5 + 4×8 + 11) / 6 = 48/6	8.00	[(11 − 5) / 6]² = 36/36	1.000
D	4	6	10	(4 + 4×6 + 10) / 6 = 38/6	6.33	[(10 − 4) / 6]² = 36/36	1.000
E	2	3	6	(2 + 4×3 + 6) / 6 = 20/6	3.33	[(6 − 2) / 6]² = 16/36	0.444
F	1	2	3	(1 + 4×2 + 3) / 6 = 12/6	2.00	[(3 − 1) / 6]² = 4/36	0.111

B, C and D have the same variance (1): their pessimistic estimate is far from the optimistic one.

EXAMPLE 3 — QUESTION 2

The network with the mean durations and the dates



EXAMPLE 3 — QUESTION 2

Earliest and latest dates: detailed calculation

Act.	Forward pass	ti
A	Start → 0	0
B	tB = tA + dA = 0 + 3	3
C	tC = tA + dA = 0 + 3	3
D	tD = tB + dB = 3 + 32/6	50/6 ≈ 8.33
E	tE = max(tC+dC ; tD+dD) = max(3 + 8 ; 50/6 + 38/6) = max(11 ; 88/6)	88/6 ≈ 14.67
F	tF = tE + dE = 88/6 + 20/6	18
End	tEnd = tF + dF = 18 + 2	20

Act.	Backward pass	t*i
End	t*End = tEnd	20
F	t*F = t*End − dF = 20 − 2	18
E	t*E = t*F − dE = 18 − 20/6	88/6 ≈ 14.67
D	t*D = t*E − dD = 88/6 − 38/6	50/6 ≈ 8.33
C	t*C = t*E − dC = 88/6 − 8	40/6 ≈ 6.67
B	t*B = t*D − dB = 50/6 − 32/6	3
A	t*A = min(t*B ; t*C) − dA = min(3 ; 40/6) − 3 = 3 − 3	0

EXAMPLE 3 — QUESTION 2

Floats, critical path and its variance

Activity	A	B	C	D	E	F
Latest start t*i	0	3	6.67	8.33	14.67	18
Earliest start ti	0	3	3	8.33	14.67	18
Float mi = t*i – ti	0	0	3.67	0	0	0
Critical?	Yes	Yes	No	Yes	Yes	Yes

CRITICAL PATH A – B – D – E – F

$$\mu = 3 + 5.33 + 6.33 + 3.33 + 2$$
$$= 120/6 = 20 \text{ weeks}$$

VARIANCE OF THE PROJECT

$$\sigma^2 = (4+36+36+16+4)/36 = 96/36$$
$$= 2.667 \rightarrow \sigma = 1.633$$

EXAMPLE 3 — QUESTIONS 3 TO 5

Choosing a launch date

$X \sim N(\mu = 20 ; \sigma = 1.633)$. $z = (x - \mu) / \sigma$, $P(X \leq x) = \Phi(z)$.

Within 22 weeks	$z = (22 - 20) / 1.633 = 1.22$	$\Phi(1.22) = 0.8888$	**88.9 %**
Within 19 weeks	$z = (19 - 20) / 1.633 = -0.61$	$\Phi(-0.61) = 1 - 0.7291 = 0.2709$	**27.1 %**
Deadline at 95 %	$\Phi(z) = 0.95 \rightarrow z = 1.645$	$x = 20 + 1.645 \times 1.633 = 22.69$	**22.7 weeks**

Decision: announce the launch for week 23 (≥ 95 % confidence), not week 20 — at week 20 the start-up has only one chance in two to be ready.

EXERCISE 2 · PROBABILISTIC NETWORK

Uncertainty on the critical path of Exercise 1

The four activities of the critical path B–D–F–H of Exercise 1 are now estimated with three values (in weeks).

Activity	a	m	b
B	4	6	8
D	3	5	7
F	1	2	3
H	2	4	6

QUESTIONS

1. Draw the network of the path B–D–F–H with the dates.
2. Compute the mean and the variance of each activity.
3. Deduce the mean duration μ and the standard deviation σ.
4. Probability that the project lasts at most 19 weeks?
5. Duration associated with a probability of 90 %?

EXERCISE 2 · SOLUTION

Path network, means and variances



Act.	a	m	b	ti = (a + 4m + b) / 6	ti	vi = [(b − a) / 6]²	vi
B	4	6	8	(4 + 4×6 + 8) / 6 = 36/6	6	[(8 − 4) / 6]² = 16/36	0.444
D	3	5	7	(3 + 4×5 + 7) / 6 = 30/6	5	[(7 − 3) / 6]² = 16/36	0.444
F	1	2	3	(1 + 4×2 + 3) / 6 = 12/6	2	[(3 − 1) / 6]² = 4/36	0.111
H	2	4	6	(2 + 4×4 + 6) / 6 = 24/6	4	[(6 − 2) / 6]² = 16/36	0.444
Σ				μ = 6 + 5 + 2 + 4	17	σ^2 = (16 + 16 + 4 + 16)/36 = 52/36	1.444

EXERCISE 2 · SOLUTION

Standard deviation and probabilities

01 **Standard deviation:** $\sigma = \sqrt{1.444} = 1.20$ weeks

02 **P(X ≤ 19):** $z = (19 - 17) / 1.20 = 1.66 \rightarrow \Phi(1.66) = 0.9515 \rightarrow 95.2\ \%$

03 **Duration at 90 %:** $\Phi(z) = 0.90 \rightarrow z = 1.28 \rightarrow x = 17 + 1.28 \times 1.20 = 18.5$ weeks

Average 17 weeks, but a 90 % commitment requires 18.5 weeks.

Caution: Exercise 1 has a SECOND critical path (B–E–G–H). If E and G were also uncertain, the real risk would be higher than computed here.

3. COST MINIMISATION

The economic dimension of a project

Shortening a project requires additional resources (staff, overtime, machines): each day gained on an activity has a direct cost. But every day saved on the project reduces the indirect costs (supervision, rent, site overheads).

The trade-off

Find the best compromise between a shorter project (less indirect cost) and the extra direct cost of the reductions.

Crash point

An activity reduced to its minimum duration has reached its crash point: it cannot be shortened any more.

REDUCTION COST PER DAY (SLOPE)

$$ki = (\text{crash cost} - \text{normal cost}) / (\text{normal duration} - \text{crash duration})$$

3. COST MINIMISATION

The procedure, step by step

01	Draw the network, compute the dates and floats, and list the critical and non-critical paths.
02	Among the CRITICAL activities, choose the cheapest reduction (lowest cost per day).
03	Reduce it by as many days as possible, but never more than the float of the nearest non-critical path.
04	Recompute the network. If several paths are critical, reduce them TOGETHER (one activity per path, or a common activity).
05	Stop when one more day costs ≥ the indirect cost saved per day, or when a critical path can no longer be shortened. Draw the cost chart.

Reducing a non-critical activity is pointless: it only increases its float.

3. EXAMPLE 4 — CRASHING A PROJECT

The data

Consider the following project. Its indirect costs amount to 1,000 TND per working day.

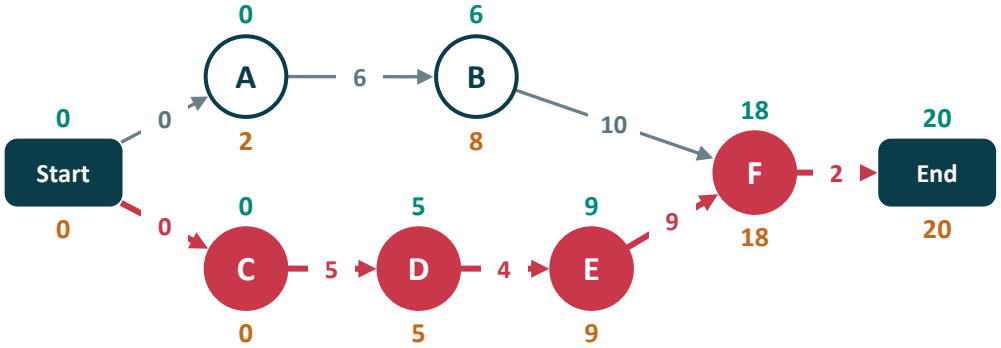
Act.	Pred.	Normal (days)	Crash (days)	Max. reduction	Cost (TND/day)
A	—	6	6	0	— (none)
B	A	10	8	2	500
C	—	5	4	1	300
D	C	4	1	3	700
E	D	9	7	2	600
F	B, E	2	1	1	800

QUESTIONS

1. Network, dates and floats with normal durations.
2. Reduce step by step; draw the new network after each reduction.
3. Optimal duration; when to stop and why.

EXAMPLE 4 — INITIAL SITUATION

Normal durations: network, dates and floats



Activity	A	B	C	D	E	F
Latest start t*i	2	8	0	5	9	18
Earliest start ti	0	6	0	5	9	18
Float mi = t*i – ti	2	2	0	0	0	0

PATHS

C–D–E–F = 5 + 4 + 9 + 2 = 20 days
critical (float 0)

A–B–F = 6 + 10 + 2 = 18 days
non-critical (float 2)

INITIAL COST
Duration: 20 days
Indirect: 20 × 1,000 = 20,000 TND

EXAMPLE 4 — BEFORE REDUCING

Why we must start with the critical activities

Wrong choice: reduce B (non-critical)

- B: 10 → 9 days, cost + 500 TND
- A–B–F: 18 → 17 days
- C–D–E–F still 20 days → project = 20 days
- Float of B: 2 → 3 days
- **Result: 500 TND spent, 0 day gained.**

The three rules

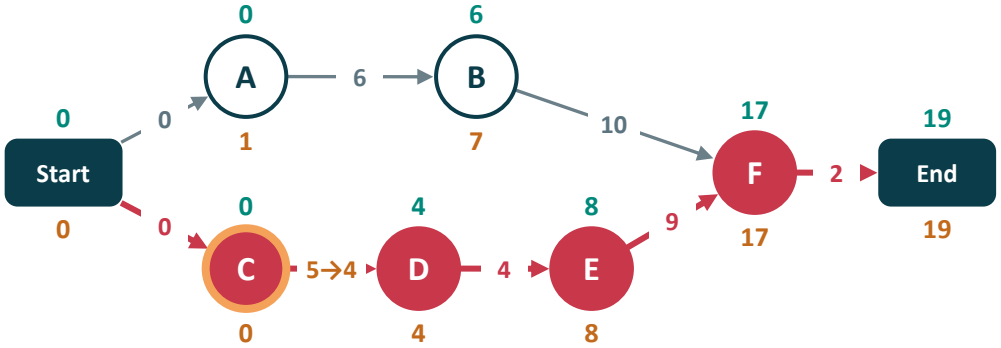
Critical only: the project lasts as long as its longest path; only shortening that path shortens the project.

Cheapest first: each day is bought at the lowest price.

Limited step: never reduce by more than the float of the nearest path (here 2 days), or the other path becomes the longest.

EXAMPLE 4 — REDUCTION 1

Reduce C by 1 day (the cheapest critical activity)



Activity	A	B	C	D	E	F
Latest start t*i	1	7	0	4	8	17
Earliest start ti	0	6	0	4	8	17
Float mi = t*i – ti	1	1	0	0	0	0

CANDIDATES (CRITICAL ONLY)

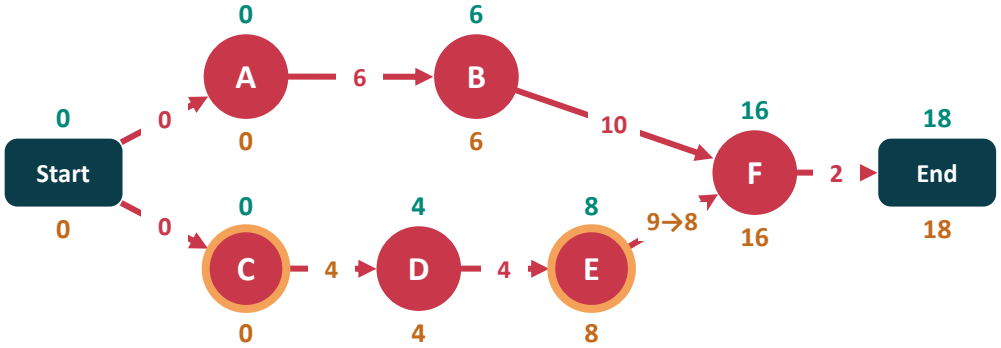
Option	TND/day	Days left
C (chosen)	300	1
E	600	2
D	700	3
F	800	1

C–D–E–F = 4+4+9+2 = 19 (critical)
A–B–F = 18 (float 1)

NEW DURATION 20 → 19 DAYS
+300 direct – 1,000 indirect = –700 TND
Cumulative saving: 700 TND

EXAMPLE 4 — REDUCTION 2

Reduce E by only 1 day: two critical paths



Activity	A	B	C	D	E	F
Latest start t*i	0	6	0	4	8	16
Earliest start ti	0	6	0	4	8	16
Float mi = t*i – ti	0	0	0	0	0	0

CANDIDATES (CRITICAL ONLY)

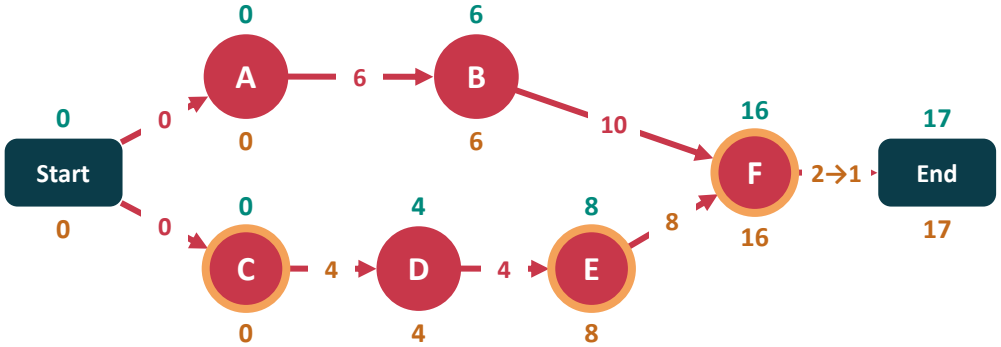
Option	TND/day	Days left
E (chosen)	600	2
D	700	3
F	800	1
C	—	0 (crash)

C–D–E–F = 4+4+8+2 = 18 (critical)
A–B–F = 18 (critical)

NEW DURATION 19 → 18 DAYS
+600 direct – 1,000 indirect = –400 TND
Cumulative saving: 1,100 TND

EXAMPLE 4 — REDUCTION 3

Both paths critical: reduce F, common to both paths



CANDIDATES (CRITICAL ONLY)

Option	TND/day	Days left
F (chosen)	800	1
B + E	1,100	1
B + D	1,200	2

C–D–E–F = 4+4+8+1 = 17 (critical)
A–B–F = 6+10+1 = 17 (critical)

Activity	A	B	C	D	E	F
Latest start t*i	0	6	0	4	8	16
Earliest start ti	0	6	0	4	8	16
Float mi = t*i – ti	0	0	0	0	0	0

NEW DURATION 18 → 17 DAYS
+800 direct – 1,000 indirect = –200 TND
Cumulative saving: 1,300 TND

EXAMPLE 4 — WHEN DO WE STOP?

At 17 days, one more day costs more than it saves

Option to gain 1 more day	Extra direct cost	Indirect saving	Net effect	Decision
B + E (one per path)	500 + 600 = 1,100	1,000	+100 TND	✗ costs more
B + D (one per path)	500 + 700 = 1,200	1,000	+200 TND	✗ costs more
F	—	—	crash reached (1 day)	✗ impossible
A	—	—	cannot be reduced	✗ impossible

Stop rule 1 — economic

Stop when the cheapest way to gain one day costs ≥ the indirect cost saved (1,000 TND/day). Here 1,100 > 1,000: STOP at 17 days.

Stop rule 2 — technical

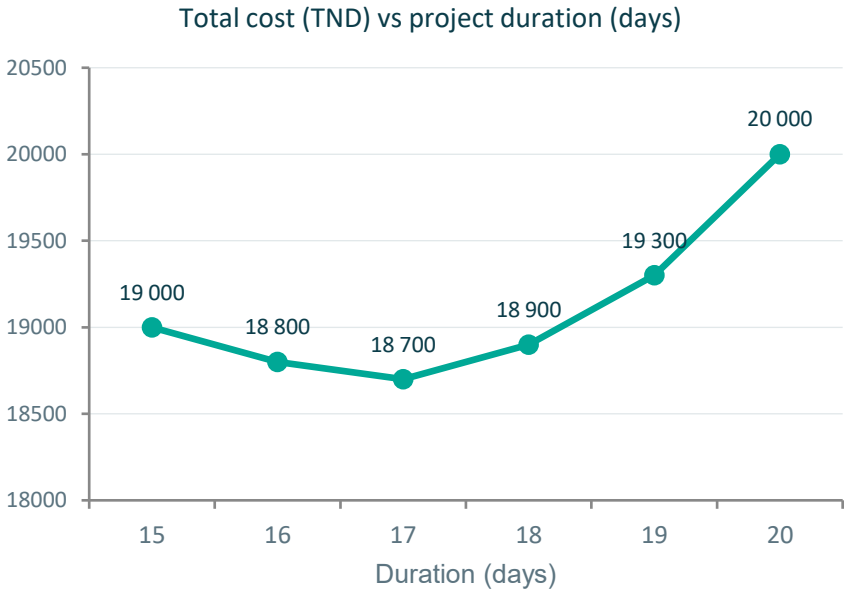
Stop anyway when a critical path is fully crashed. If we continued: 16 days, then 15 days; then A–B–F = 6 + 8 + 1 = 15 cannot shrink.

EXAMPLE 4 — OPTIMISATION CHART

Total cost is minimum at 17 days

Duration	Reduction	Extra direct	Indirect	Total*
20	—	0	20,000	20,000
19	C –1	300	19,000	19,300
18	E –1	900	18,000	18,900
17	**F –1**	**1,700**	**17,000**	**18,700**
16	B, E –1	2,800	16,000	18,800
15	B, D –1	4,000	15,000	19,000

** Total = extra direct + indirect cost, in TND (the normal direct cost is the same for every duration).*



Thank you for your attention

Next session: tutorial on potential networks, probabilities and crashing. Bring the standard normal table.

Potential network

Earliest / latest dates

Floats & critical path

Probability & crashing

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